Discrete Mathematics and graph Theory
Chapter Two: Recurrence relation

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Out lines

Topics

In this chapter we will discuss the following topics:

2.1. The notion of sequences.

2.2. Recursive definition.

2.3. Recurrence relations.

2.4. Homogeneous recurrence relations.

2.5. Non homogeneous recurrence relations.

The notion of a sequence

Definition

A sequence is a function whose domain is the subset of integers $n \geq n_0$ (usually 1).

Notation: A sequence is denoted by a_n, f_n or $f(n)$, and is written as $\{a_n\}_1^n$ or simply $\{a_n\}$.

Descriptions of sequences

A sequence can be described by

1. enumerating the first few terms(enumeration method)

Example

The sequence $\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{16}, \cdots\}$.

2. supplying the general term (Explicit Method).

Example

The sequence $a_n = \frac{1}{n}, n \in \mathbb{N}$.

3. recursive definition.

Example

A Fibonacci sequence $a_n = a_{n-1} + a_{n-2}$, $a_0 = 1$, $a_2 = 1$.

There are many counting problems that cannot be solved easily using the methods discussed in the previous chapter. In order to solve such problems we have to find relationships between consecutive terms. The rule for finding terms from those that precede them is called a recurrence relation.

Models of recurrence relations

Recurrence relations have applications in computer science(such as algorithm analysis), in business(such as finding compound interest), in engineering(for dynamical system and optimization), in biology(population dynamics). The following are some of the models.

Example (Compound interest)

Suppose that a person deposits \$10,000 in a saving account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after n years?

Let P_n denote the amount in the account after n years. Here, the amount in the account after n years equals the amount in the account after $n-1$ years plus interest in the n^{th} year. This model is given by the following recurrence relation.

$$P_n = P_{n-1} + 0.11P_{n-1} = 1.11P_{n-1}, n \geq 1, \text{ where } P_0 = \$10,000$$

Example (Reproduction of Rabbits and Fibonacci numbers)

A young pair of rabbits(one of each sex) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. What will be the recurrence relation for the number of pairs rabbits on the island after n months, assuming no rabbits ever die?

Here the model which represent the number of population of rabbits is a recurrence relation given by

$$f_n = f_{n-1} + f_{n-2}, \text{ for } n \geq 2, \text{ with initial condition } f_0 = 1, f_1 = 1$$

This model is called Fibonacci sequence.

Example

The number of bacteria in a colony doubles every hour. If the colony begins with 5 bacteria, how many will be present in n hours?

The model for this problem is given by

$$a_n = 2a_{n-1}, a_0 = 5$$

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Recursive definition

A technique of defining an algorithm, a set or a function in terms of itself by:

(i) Giving a rule for finding present and future values from earlier or prior values.

(ii) Specifying one or more starting values to activate the rule mentioned in (i)

An algorithm, a set or a function stated in terms of these two conditions is called a recursively defined algorithm, set or function, respectively.
Note that a recursive definition is **well defined** if and only if it satisfies conditions (i) and (ii) above.

Example

- ☞ The sequence defined by $a_{n+1} = 3a_n$ where, $n \geq 0$ and $a_0 = 5$ is a well defined geometric sequence.
- ☞ The sequence defined by $a_{n+1} = 3a_n$ where, $n \geq 0$ is not well defined since it not unique.

Definition

A recurrence relation (difference equation) for a sequence $\{a_n\}$ is an equation that express a_n in terms of one or more of the previous terms of a sequence; namely $a_0, a_1, \cdots, a_{n-1}$ for all integers $n \geq n_0$.

- ☛ Initial condition: are one or more starting terms of the sequence.

Example

A sequence
$f_n = f_{n-1} + f_{n-2},$ for $n \geq 2,$ with initial condition $f_0 = 1, f_1 = 1$, is the recurrence relation.

Solution of a Relation:

Definition

A sequence $a_n = f(n)$ is called a solution of a recurrence relation if it satisfies the recurrence relation.

Example

1. For the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$, $n \geq 2$, the following are solutions.

$$a_n = 3n \text{ for all } n \geq 0.$$
$$a_n = 5 \text{ for all } n \geq 0.$$

Exercise

1. *Show that* $a_n = 2^{n+1} - 1$ *is a solution of the recurrence relation* $a_n = 3a_{n-1} - 2a_{n-2}$, $n \geq 3$.
2. *Show that* $f_n = 2(-4)^n + 3$ *is a solution of the recurrence relation* $f_n = -3f_{n-1} + 4f_{n-2}$, $n \geq 3$.
3. *Consider the recurrence relation*

$$a_n = 2a_{n-1} - a_{n-2}(n \geq 2).$$

Which of the following are solutions?
(a) $a_n = 3n$ *(b)* $a_n = 2n$ *(c)*$a_n = 5$

Linear Recurrence Relation(LRR)

Definition

A recurrence relation of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = f(n), \qquad (1)$$

$(c_0, c_k \neq 0; 1 \leq k \leq n)$ is called a linear recurrence relation.

Remark

- *If $c_0, c_1, c_2, \cdots c_k$ are constants, it is called recurrence relation with constant coefficients.*
- *If $f(n) = 0$ for all n, the recurrence relation is called homogeneous. Otherwise, it is called non homogeneous.*

Example

1. The recurrence relation $a_n = a_{n-1} + a_{n-2}$, $n \geq 2$ is linear.

Example

2. The recurrence relation $a_n = a_{n-1} - 3a_{n-2}^2$, $n \geq 2$ is non linear.

Order of the recurrence relation:

The order of the recurrence relation is the difference between the highest and lowest indices of the terms in the relation.
It can also be defined as the number of preceding terms that appear in the recurrence relation.

Example

Classify the following recurrence relations as homogeneous or non homogeneous, and determine their order.

(a) $a_n = 2a_{n-1} - 3n, n \geq 1$

(b) $a_n = a_{n-1} + a_{n-2}, n \geq 2.$

(c) $a_n - a_{n-2} = 0, n \geq 2.$

(d) $a_{n+1} - 5a_n + 4a_{n-1} - 6a_{n-2} = 3n + 1, n \geq 3.$

General Solution:

Definition

The general solution is a solution $a_n = f(n)$ that represents the entire family of possible solutions for a recurrence relation, expressed with arbitrary constants.

Unique solution

Definition

The unique solution is a specific formula $a_n = f(n)$ that satisfies the Recurrence Relation with a particular set of initial conditions.

Example

1. $an = c_1(-3)^n + c_2 n(-3)^n$ is the general solution for the recurrence relation $a_n = -6a_{n-1} - 9a_{n-2}$.
2. $an = (2 + n)(-3)^n$ is the unique solution for the recurrence relation $a_n = -6a_{n-1} - 9a_{n-2}$ with initial conditions $a_0 = 2$, and $a_1 = -9$.

Solving Linear Homogeneous RRs

The general form of the k^{th} order linear homogeneous recurrence relation is given by

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = 0 \qquad (2)$$

where $c_k \neq 0$, $n \geq k$.

Properties

The solutions of a homogeneous linear RRs have two basic properties.

- **(i)** A non zero constant multiple of a solution of (2) is also a solution.
- **(ii)** The sum of two or more solutions of (2) is also a solution.

Solving Linear Homogeneous recurrence relations with constant coefficients(LHRRWCC)

(a) First order.

The general form of the first order linear homogeneous recurrence relation is given by

$$c_0 a_n + c_1 a_{n-1} = 0 \tag{3}$$

Note

The general solution of the first order recurrence relation (3) is of the form $a_n = \alpha r^n$, $n \geq 0$, where r is determined from the linear equation(called the characteristic or auxiliary equation) given by

$$c_0 r + c_1 = 0$$

Example

Solve the following recurrence relations.

(a) $a_n = 4a_{n-1}, n \geq 1$ with $a_1 = 3$.

(b) $2a_{n+1} - a_n = 0, n \geq 0$.

(c) $4a_n - 5a_{n-1} = 0$.

(d) $3a_n - 4a_{n-1} = 0, n \geq 1$ with $a_1 = 5$

(b) The second order.

The general form of the second order linear homogeneous recurrence relation is given by

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0, n \geq 2 \tag{4}$$

where $c_0, c_2 \neq 0$.

Note

The solution of the second order recurrence relation (4) is of the form $a_n = r^n$, $n \geq 0$, where r is determined from the quadratic equation(called the characteristic or auxiliary equation)given by

$$c_0 r^2 + c_1 r + c_2 = 0$$

Cases for the solutions

The root(s) of the characteristic equation $c_0 r^2 + c_1 r + c_2 = 0$ is solved using a quadratic formula $r = \dfrac{-c_1 \pm \sqrt{c_1^2 - 4c_0c_2}}{2c_0}$, and there are three cases.

Case 1. If r_1 and r_2 are then two roots, then $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, $n \geq 0$ is the general solution.

Case 2. If $r_1 = r_2$ is the only root, then $a_n = \alpha_1 r_1^n + \alpha_2 n r_1^n$, $n \geq 0$ is the general solution.

Case 3. If r_1 and r_2 are then two complex roots, then $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, $n \geq 0$ is the general complex solution.

Example

Solve the following recurrence relations.

(a) $a_n + 5a_{n-1} + 6a_{n-2}, n \geq 2$ with $a_0 = 3$, and $a_1 = 5$.

(b) $a_n = 6a_{n-1} - 9a_{n-2}, n \geq 2$ with initial conditions $a_0 = 1$, and $a_1 = 6$.

(c) $a_n - 4a_{n-1} + 4a_{n-2}, n \geq 2$ with initial conditions $a_0 = 2$, and $a_1 = 5$.

(d) $a_n + 4a_{n-2} = 0, n \geq 2$.

(c) The k^{th} order recurrence relations.

The general form of the k^{th} order linear homogeneous recurrence relation is given by

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = 0, n \geq k \qquad (5)$$

where $c_0, c_k \neq 0$.

Note

The solution of the second order recurrence relation (4) is of the form $a_n = r^n$, $n \geq 0$, where r is determined from the equation(called the characteristic or auxiliary equation)given by

$$c_0 r^k + c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_{k-1} r + c_k = 0$$

General Steps

The following are steps to solve the homogeneous recurrence relations.

1. Find the corresponding characteristic equation.
2. Determine the roots of the characteristic equation.
3. Write the general solution.
4. Use initial conditions (if given) to find the unique solution.

Example

Solve the following recurrence relations.

(a) $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}, n \geq 2$ with $a_0 = 3$, $a_1 = 5$ and $a_2 = 15$.

(b) $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}, n \geq 3$ with initial conditions $a_0 = 1$,$a_1 = -2$, and $a_2 = 4$.

(c) $a_n = 3a_{n-2} - 2a_{n-3}, n \geq 3$ with initial conditions $a_0 = a_1 = 1$, and $a_2 = 4$.

Non homogeneous linear recurrence relations

The general form of the k^{th} order non linear recurrence relation is given by

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = f(n),$$

$(c_0, c_k \neq 0;)$, where $f(n) \neq 0$ for some n.

Note

The general solution of the linear nonhomogeneous recurrence relation with constant coefficients is given by

$$a_n = a_n^{(h)} + a_n^{(p)},$$

where $a_n^{(h)}$ is the general solution of the associated homogeneous recurrence relation, and $a_n^{(p)}$ is its particular solution.

How to find $a_n^{(p)}$?

Method of Undetermined Coefficients:

☞ There is no general technique for determining the particular solution $a_n^{(p)}$ that fits a given difference equation. However, when the function $f(n)$ has a certain form, we can inductively suggest its form (i.e., the form of $a_n^{(p)}$, in line with the nature of $f(n)$). The following are guide-lines in choosing a suitable particular solution $a_n^{(p)}$.

1. If $f(n)$ is a polynomial function of degree k, the particular solution is of the form

$$a_n^{(p)} = A_0 + A_1 n + A_2 n^2 + \cdots + A_k n^k.$$

Example

Solve the following non homogeneous recurrence relation:

(a) $a_{n-} - 7a_{n-1} + 12a_{n-2} = 1$, $n \geq 2$ with boundary conditions: $a_0 = 0$ and $a_1 = 1$.

(b) $a_n - 3a_{n-1} = 3n - 2$, $n \geq 1$ and $a_0 = 1$.

(c) $a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$.

2. If $f(n) = \beta r^n$, where β and r are constants, we have the following cases.

(a) If r is not the root of the characteristic equation of the associated homogeneous recurrence relation, the particular solution has the form $a_n^{(p)} = Ar^n$, for some constant A.

(b) If r is the root if r is a root of multiplicity $m \geq 1$ for the homogeneous case of the characteristic equation of the associated homogeneous recurrence relation, the particular solution has the form $a_n^{(p)} = An^m r^n$, for some constant A.

Example

Solve the following non homogeneous recurrence relation:

- **(a)** $a_n – 2a_{n-1} = 5(4)^n$, $n \geq 1$, $a_0 = 2$.
- **(b)** $a_n – 4a_{n-1} + 4a_{n-2} = 5(4)^n$, $n \geq 2$, $a_0 = 2$, $a_1 = 5$.
- **(c)** $a_n – 6a_{n-1} + 8a_{n-2} = n(4)^n$, $n \geq 2$ with its initial conditions $a_0 = 8$ and $a_1 = 22$.
- **(d)** $a_n – 4a_{n-1} + 4a_{n-2} = 3(2)^n$, $n \geq 2$, $a_0 = 2$, $a_1 = 5$.

Exercise

1. *Solve the following non homogeneous recurrence relation:*
   (a) $a_n - 4a_{n-1} + 4a_{n-2} = 2n + 3 - (4)^n$, $n \geq 2$, $a_0 = 2$, $a_1 = 5$.
   (b) $a_n - 6a_{n-1} + 8a_{n-2} = n^2 + 2^n$, $n \geq 2$ *with its initial conditions* $a_0 = 8$ *and* $a_1 = 22$.
   (d) $a_n - 4a_{n-2} = 3 - (2)^n$, $n \geq 2$, $a_0 = 3$, $a_1 = 8$.

Thank You!